

Micro-StudioPLC Specification

Open Standard for Inter Controller communications.

This specification covers the serial communication exchanges between any type of Controllers (embedded devices, computers, SOC/SBC, etc ...).

Generally, MSPLC Software supports Application Class devices and kRT kernel supports any controller with at least 1k of Flash and 100 bytes of Ram/Sram, a timer and a serial communication port.

The present specifcation is OS/Device agnostic, any device corresponding to the present document can be interconnected and accessed through the MSPLS protocol.

V1.0 of the protocol defines a Kernel and User runtime application on a controller. The protocol defines 135 base commands and supports arbitrary frames/extended commands.

The present Specification provides the tools to standardize peripheral access, for example

- A sensor (temperature, hygrometry, accelerometer, ...)
- A peripheral (LCD Screen, HMI, ...)
- A Generic MCU (microcontroller with Analog and Digital GPIOs)
- An embedded controller (Raspberry, BeagleBoard, Arduino, ...)
- An industrial controller (Open source and Arduino bases PLCs)

The Device manufacturers implements some of the specification commands as needed and documents their Usage through the Datasheet/Man pages.

For example a miniman working device such as temperature sensor may be read through SPI Bus with the following command *g0ra*: Reads Channel 0 (temperature) as 16 bit signed integer.

In this example, the Thermometer CI only supports 1 command and 1 channel. Readback the Analog channel 0 provides a direct reading of the temperature as 16-bit signed integer.

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Definitions

A *Controller* is a generic term for any device which can be addressed via a serial bus.

It can range from simple microcontrollers such as 8 Bits MCU with I2C/SPI to Modbus devices and custom or Open-Source PLC controllers.

Command list

Kernel Commands:

For complex devices, Kernel commands to control and monitor the state of the controller can be implemented.

Msg	Xch Spec	Hex	Description
M1	skst, stopkst	0x01	Stop K-ST (Kernel Space scheduler)
M2	rkst, runkst	0x02	Run K-ST (Kernel Space scheduler)
M4	sys, systat	0x04	Read-back systat then reset min/max
M8	clr, clear	0x08	Clear Fault and Alarm Words
M16	sust, stopust	0x10	Stop U-ST (User Space scheduler)
M32	rust, runust	0x20	Run U-ST (User Space scheduler)
M64	sta, status	0x40	Return Status, Alarm and Fault words

User Commands:

Generic set of commands to Configure, Read and Write data from/to a controller

Msg	Xch Spec	Hex	Description
M128	g[n]ci	0x80	Set GPIO [n] as digital input
M129	g[n]co	0x81	Set GPIO [n] as digital output
M130	g[n]ca	0x82	Set GPIO [n] as analog input. Server responds with PWM resolution in bits.
M131	g[n]cp	0x83	Set GPIO [n] as PWM output. Server responds with PWM resolution in bits.
M132			
M133			
M134	g[n]cs+data	0x86	Configure special. User defined.
M135	g[n]cr	0x87	Read-back GPIO [n] Configuration
M136	g[n]cod	0x88	Configure GPIO [n] Open Drain (High Z)
M137	g[n]cpu	0x89	Configure GPIO [n] Pull Up
M138	g[n]cpd	0x8a	Configure GPIO [n] Pull Down